Polynomial Continued Fractions: Proved Families, Casoratian Series Identities, and 482 New Irrational Constants

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Abstract

We present two infinite parametric families of polynomial continued fraction (PCF) identities sharing the common denominator $\beta(n) = 3n+1$. The *Logarithmic Ladder*, $\mathrm{PCF}(-kn^2,\ (k+1)n+k) = 1/\ln(k/(k-1))$ for all real $k > 1$, is **proved** by exhibiting closed-form convergents that reduce to the Taylor series $-\ln(1-x)/x$. The *Pi Family*, $\mathrm{PCF}(-n(2n-(2m+1)),\ 3n+1) = 2^{2m+1}/(\pi\binom{2m}{m})$, is **proved** at the base case $m = 0$ $(2/\pi)$ via the arcsin series, and at $m = 1$ $(4/\pi)$ via a novel Casoratian (discrete Wronskian) argument that yields a new rapidly convergent series identity

$$\frac{\pi}{4} = 1 - \sum_{k=1}^{\infty} \frac{k!}{(2k-1)!!\,(k^2+3k+1)(k^2+k-1)},$$

with exponential convergence rate $O(2^{-N}/N^{7/2})$.

Our main result unifies both integer and half-integer specialisations into a single continuous identity: for all real $m > -\frac{1}{2}$, the PCF converges to $2\,\Gamma(m+1)/(\sqrt{\pi}\,\Gamma(m+\frac{1}{2}))$. This is established via a raising operator mapping solutions between adjacent recurrences, combined with Poincaré growth analysis. The Pi Family exhibits a parity phenomenon: odd parameters yield transcendental multiples of $1/\pi$, while even parameters collapse to Wallis-type rationals.

We prove that the $m = 1$ CF does not arise from any ${}_2F_1$ via the Gauss continued-fraction theorem; the associated series lies in the class ${}_2F_3$ over $\mathbb{Q}(\sqrt{5})$. A comparison with Brouncker's classical CF for $4/\pi$ reveals dramatic structural differences: exponential convergence $O(2^{-n})$ versus $O(1/n)$, with a factor of 10^4 at $n = 8$.

Finally, we catalogue 482 provably irrational constants from quadratic-denominator generalised continued fractions, all certified by ball arithmetic to 1 500+ digits, with systematic PSLQ searches against 15 standard constants finding no algebraic relations.

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Contents

1 Introduction

A *polynomial continued fraction* (PCF) with numerator polynomial $\alpha(n)$ and denominator polynomial $\beta(n)$ is

$$v = \beta(0) + \cfrac{\alpha(1)}{\beta(1) + \cfrac{\alpha(2)}{\beta(2) + \cfrac{\alpha(3)}{\beta(3) + \cdots}}}. \tag{1}$$

The convergents $C_n = p_n/q_n$ are computed via the forward recurrence

$$p_n = \beta(n)\, p_{n-1} + \alpha(n)\, p_{n-2}, \qquad q_n = \beta(n)\, q_{n-1} + \alpha(n)\, q_{n-2}, \tag{2}$$

with $p_{-1} = 1$, $p_0 = \beta(0)$, $q_{-1} = 0$, $q_0 = 1$.

Such objects encode deep arithmetic: the Gauss continued fraction expresses ${}_2F_1$ ratios [?, ?], and Apéry's celebrated proof of irrationality of $\zeta(3)$ [?] hinges on a PCF with degree-six numerator. The Ramanujan Machine project [?, ?] has revived systematic search for PCF representations of mathematical constants.

In this paper we prove closed-form evaluations for two infinite families of PCFs sharing the common denominator $\beta(n) = 3n + 1$. The first, the *Logarithmic Ladder* (§??), produces $1/\ln(k/(k-1))$ for all real $k > 1$. The second, the *Pi Family* (§§??–??), converges to $2\,\Gamma(m+1)/(\sqrt{\pi}\,\Gamma(m+\frac{1}{2}))$ for all real $m > -\frac{1}{2}$, unifying π-formulas at integer m with exact Wallis rationals at half-integer m into a single continuous Gamma-function identity (Theorem ??). For the $m = 1$ specialisation $S = 4/\pi$, we give a self-contained proof via the discrete Wronskian (Casoratian), yielding a new series for $\pi/4$ (§??). We establish that this CF does not arise from any ${}_2F_1$ via the Gauss theorem (§??), compare it with Brouncker's classical CF (§??), and present a catalogue of 482 new irrational constants from quadratic GCFs (§??).

We emphasise the asymmetry of our results: the Logarithmic Ladder and the base cases $m = 0, 1$ of the Pi Family are fully proved; the Gamma identity for general m is established conditionally on a ratio formula (Conjecture ??) supported by 500+ digits of evidence.

2 The Logarithmic Ladder

Theorem 2.1 (Logarithmic Ladder)**.** *For all real $k > 1$, the PCF with $\alpha(n) = -kn^2$ and $\beta(n) = (k+1)n + k$ converges to $1/\ln(k/(k-1))$.*

Proof. **Claim.**

$$p_n = (n+1)!\, k^{n+1}, \qquad q_n = (n+1)! \sum_{j=0}^{n} \frac{k^{n-j}}{j+1}. \tag{3}$$

Base case. $p_0 = \beta(0) = k = 1! \cdot k^1$ and $q_0 = 1 = 1! \cdot k^0/1$.

Induction for p_n. Assume $p_{n-1} = n!\, k^n$ and $p_{n-2} = (n-1)!\, k^{n-1}$.

$$\begin{aligned}\beta(n)\, p_{n-1} + \alpha(n)\, p_{n-2} &= \big((k+1)n + k\big) n!\, k^n - kn^2 (n-1)!\, k^{n-1} \\ &= n!\, k^n \big((k+1)n + k - n\big) = n!\, k^n \cdot k(n+1) = (n+1)!\, k^{n+1}.\end{aligned}$$

Induction for q_n. Write $S_n = \sum_{j=0}^{n} k^{n-j}/(j+1)$ and note $S_n = k\, S_{n-1} + 1/(n+1)$. Assume $q_{n-1} = n!\, S_{n-1}$ and $q_{n-2} = (n-1)!\, S_{n-2}$. Using $S_{n-1} = k\, S_{n-2} + 1/n$:

$$\begin{aligned}\beta(n)\, q_{n-1} + \alpha(n)\, q_{n-2} &= \big((k+1)n+k\big)\, n! \big(k\, S_{n-2} + \tfrac{1}{n}\big) - kn^2\, (n-1)!\, S_{n-2} \\ &= S_{n-2} \cdot k^2 (n+1)\, n! + (n-1)! \big(kn + k + n\big).\end{aligned}$$

Meanwhile, $(n+1)!\, S_n = k^2 (n+1)\, n!\, S_{n-2} + k(n+1)(n-1)! + n!$. Since $(n-1)!(kn+k+n) = k(n+1)(n-1)! + n!$, the two expressions are equal.

Limit. The closed forms (??) are polynomial identities in k, valid for all real k. The convergent ratio is $C_n = k/\sum_{j=0}^{n} (1/k)^j/(j+1)$. For $|1/k| < 1$, the series $\sum_{j=0}^{\infty} x^j/(j+1) = -\ln(1-x)/x$ converges at $x = 1/k$, giving $\sum (1/k)^j/(j+1) = k \ln(k/(k-1))$. Therefore $\lim C_n = k/(k \ln(k/(k-1))) = 1/\ln(k/(k-1))$. □

Remark 2.2. The theorem holds for all real $k > 1$, not just integers. This has been verified numerically for 15 non-integer values in $[1.5, 100]$ to 90–120 matching digits.

Table 1: Logarithmic Ladder: first instances

k	$\alpha(n)$	$\beta(n)$	Value
2	$-2n^2$	$3n+2$	$1/\ln 2$
3	$-3n^2$	$4n+3$	$1/\ln(3/2)$
4	$-4n^2$	$5n+4$	$1/\ln(4/3)$
5	$-5n^2$	$6n+5$	$1/\ln(5/4)$
φ	$-\varphi\, n^2$	$(\varphi{+}1)n{+}\varphi$	$1/\ln(\varphi/(\varphi{-}1))$
k	$-kn^2$	$(k{+}1)n{+}k$	$1/\ln(k/(k{-}1))$

3 The Pi Family

3.1 Base case: $2/\pi$

Theorem 3.1 (Pi Family, $m = 0$). *The PCF with $\alpha(n) = -n(2n{-}1)$ and $\beta(n) = 3n+1$ converges to $2/\pi$.*

Proof. **Claim.**

$$p_n = (2n{+}1)!!, \qquad q_n = (2n{+}1)!! \sum_{j=0}^{n} \frac{j!}{(2j{+}1)!!}. \tag{4}$$

Base case. $p_0 = 1 = (1)!!$ and $q_0 = 1 = 1 \cdot (0!/1!!)$.

Induction for p_n. Assume $p_{n-1} = (2n{-}1)!!$ and $p_{n-2} = (2n{-}3)!!$. Using $(2n{-}1)!! = (2n{-}1)(2n{-}3)!!$:

$$\begin{aligned}\beta(n)\, p_{n-1} + \alpha(n)\, p_{n-2} &= (3n{+}1)(2n{-}1)!! - n(2n{-}1)(2n{-}3)!! \\ &= (2n{-}3)!!\big[(3n{+}1)(2n{-}1) - n(2n{-}1)\big] \\ &= (2n{-}3)!!(2n{-}1)(2n{+}1) = (2n{+}1)!!.\end{aligned}$$

Induction for q_n. Write $T_n = \sum_{j=0}^{n} j!/(2j{+}1)!!$. Note $T_{n-1} = T_{n-2} + (n{-}1)!/(2n{-}1)!!$. With inductive hypotheses $q_{n-1} = (2n{-}1)!!\, T_{n-1}$ and $q_{n-2} = (2n{-}3)!!\, T_{n-2}$:

$$\beta(n)\, q_{n-1} + \alpha(n)\, q_{n-2} = (2n{-}1)!!\big[(3n{+}1)\, T_{n-1} - n\, T_{n-2}\big].$$

Substituting $T_{n-1} = T_{n-2} + (n{-}1)!/(2n{-}1)!!$:

$$(3n{+}1)\, T_{n-1} - n\, T_{n-2} = (2n{+}1)\, T_{n-2} + \frac{(3n{+}1)(n{-}1)!}{(2n{-}1)!!}.$$

The identity reduces to $(3n{+}1)(n{-}1)! = (2n{+}1)(n{-}1)! + n!$, which holds since $n! = n(n{-}1)!$.

Limit. $C_n = p_n/q_n = 1/T_n$. The identity [?, §5.2]

$$\frac{\pi}{2} = \sum_{j=0}^{\infty} \frac{j!}{(2j{+}1)!!} = \sum_{j=0}^{\infty} \frac{2^j (j!)^2}{(2j{+}1)!}, \tag{5}$$

derivable from $\arcsin(1) = \pi/2$, gives $\lim T_n = \pi/2$ and $\lim C_n = 2/\pi$. □

3.2 Parity phenomenon

Theorem 3.2 (Parity). *For even $c = 2\ell$, the PCF with $\alpha(n) = -n(2n{-}c)$ and $\beta(n) = 3n+1$ evaluates to an exact rational.*

Proof. $\alpha(\ell) = -\ell(2\ell - 2\ell) = 0$, so the CF terminates at depth $\ell{-}1$. □

Table 2: Pi Family: even parameters (exact rationals)

c	Value	Formula
2	1	1
4	3/2	$\binom{2}{1}/2$
6	15/8	$\binom{4}{2}/2^3$
8	35/16	$\binom{6}{3}/2^4$
10	315/128	$\binom{8}{4}/2^7$
12	693/256	$\binom{10}{5}/2^8$

3.3 Odd parameters and the binomial recurrence

Conjecture 3.3 (Ratio Formula). *Define* $\mathrm{val}(m)$ *as the limit of the PCF with* $\alpha(n) = -n(2n-(2m+1))$ *and* $\beta(n) = 3n+1$. *Then* $\mathrm{val}(m+1) = \mathrm{val}(m) \cdot 2(m+1)/(2m+1)$.

Remark 3.4. The recurrence is verified numerically to 500+ matching digits for $m = 0, 1, \ldots, 10$ (and at half-integer values $m = 0.5, \ldots, 10.5$). It is consistent with the closed form $\mathrm{val}(m) = 2^{2m+1}/(\pi\binom{2m}{m})$, since $\binom{2m+2}{m+1} = \binom{2m}{m}\cdot 2(2m+1)/(m+1)$ gives ratio $4\binom{2m}{m}/\binom{2m+2}{m+1} = 2(m+1)/(2m+1)$.

Corollary 3.5 (Full Pi Family). *Assuming Conjecture* **??**, *for all* $m \geq 0$ *the PCF converges to* $2^{2m+1}/(\pi\binom{2m}{m})$.

Table 3: Pi Family: odd parameter $c = 2m+1$

m	c	$\alpha(n)$	$\mathrm{val}(m) \cdot \pi$	$\mathrm{val}(m)$
0	1	$-n(2n-1)$	2	$2/\pi$
1	3	$-n(2n-3)$	4	$4/\pi$
2	5	$-n(2n-5)$	16/3	$16/(3\pi)$
3	7	$-n(2n-7)$	32/5	$32/(5\pi)$
4	9	$-n(2n-9)$	256/35	$256/(35\pi)$

4 Convergent Closed Forms

Theorem 4.1 ($m = 0$ convergent numerators). *For the Pi Family with* $m = 0$, $p_n = (2n+1)!! = (2n-1)!! \cdot (2n+1)$.

Proof. Base cases: $p_0=1$, $p_1=3$, $p_2=15$. Induction: assume $p_k = (2k+1)!!$ for $k < n$. The recurrence $p_n = (3n+1)p_{n-1} - n(2n-1)p_{n-2}$ with $P(n) = 2n+1$ reduces to the identity $(2n-1)(2n-3)P(n) = (3n+1)(2n-3)P(n-1) - n(2n-1)P(n-2)$ after dividing by $(2n-3)!!$. Expanding both sides yields $0 = 0$. □

Theorem 4.2 ($m = 1$ convergent numerators). *For the Pi Family with* $m = 1$ *(*$\alpha(n) = -n(2n-3)$, $\beta(n) = 3n+1$*),*

$$p_n = (2n-1)!!\,(n^2 + 3n + 1). \tag{6}$$

The polynomial $h_n = n^2+3n+1$ *(OEIS [?]) has discriminant* 5 *and roots* $(-3 \pm \sqrt{5})/2$ *related to the golden ratio.*

Proof. After the substitution $p_n = (2n-1)!!\,u_n$ (using $(2n-1)!! = (2n-1)(2n-3)!!$), the recurrence reduces to

$$(2n-1)\,u_n = (3n+1)\,u_{n-1} - n\,u_{n-2}. \tag{7}$$

Substituting $u_n = n^2{+}3n{+}1$, $u_{n-1} = n^2{+}n{-}1$, $u_{n-2} = n^2{-}n{-}1$:

$$\begin{aligned}\text{RHS} &= (3n{+}1)(n^2{+}n{-}1) - n(n^2{-}n{-}1)\\ &= 2n^3 + 5n^2 - n - 1 = (2n{-}1)(n^2{+}3n{+}1) = \text{LHS}.\end{aligned}$$

□

Conjecture 4.3 (General m)**.** *For all $m \geq 0$ and fixed n, the quantity $R(n,m) = p_n(m)/(2n{-}1)!!$ is a polynomial in m of degree $\lfloor n/2 \rfloor$.*

This is verified for $m = 0, \ldots, 12$ and $n \leq 20$. For $m \geq 2$, $R(n,m)$ is rational with denominators dividing the product of odd primes $\leq 2n{-}1$.

5 The Continuous Gamma Identity

The results of §§**??**–**??** are special cases of a single continuous identity. Its extension beyond the base cases $m = 0, 1$ is conditional on Conjecture **??**.

Theorem 5.1 (Gamma Function Representation)**.** *The formula below is proved unconditionally for $m = 0$ and $m = 1$. Assuming Conjecture **??**, for all real $m > -\frac{1}{2}$, the PCF with $\alpha(n) = -n(2n{-}2m{-}1)$ and $\beta(n) = 3n + 1$ converges to*

$$\text{val}(m) = \frac{2\,\Gamma(m{+}1)}{\sqrt{\pi}\,\Gamma(m{+}\frac{1}{2})}.$$

Proof. We prove the identity for $m = 0$ and $m = 1$ by a purely algebraic argument, then extend to all m via the Wallis reduction formula and Conjecture **??** (supported by 500+ digit evidence).

Step 1: The raising operator. Define $L(f)_n = (n{+}2)\, f_n - (n{+}1)^2\, f_{n-1}$.

Lemma 5.2. *If f satisfies the $m{=}0$ recurrence $f_n = (3n{+}1)f_{n-1} - n(2n{-}1)f_{n-2}$, then $g = L(f)$ satisfies the $m{=}1$ recurrence $g_n = (3n{+}1)g_{n-1} - n(2n{-}3)g_{n-2}$.*

Proof of Lemma. Write $g_{n-1} = (n{+}1)f_{n-1} - n^2 f_{n-2}$ and $g_{n-2} = n\, f_{n-2} - (n{-}1)^2 f_{n-3}$. Expanding $(3n{+}1)g_{n-1} - n(2n{-}3)g_{n-2}$ and eliminating f_{n-3} via the $m{=}0$ recurrence at index $n{-}1$ yields coefficients

$$\begin{aligned}&\text{coeff. of } f_{n-1} : (3n^2{+}4n{+}1) - n(n{-}1) = 2n^2{+}5n{+}1,\\ &\text{coeff. of } f_{n-2} : -n^2(5n{-}2) + n(n{-}1)(3n{-}2) = -n(2n^2{+}3n{-}2),\end{aligned}$$

matching $g_n = (n{+}2)[(3n{+}1)f_{n-1} - n(2n{-}1)f_{n-2}] - (n{+}1)^2 f_{n-1}$ identically. □

Since L is linear, it maps the solution space $\mathcal{S}_0$ of the $m{=}0$ recurrence into $\mathcal{S}_1$. Checking initial values:

$$\begin{aligned}&L(P^{(0)})_0 = 2\cdot 1 - 1\cdot 1 = 1 = P_0^{(1)}, \quad L(P^{(0)})_1 = 3\cdot 3 - 4\cdot 1 = 5 = P_1^{(1)},\\ &L(Q^{(0)})_0 = 2\cdot 1 - 1\cdot 0 = 2 = 2\,Q_0^{(1)}, L(Q^{(0)})_1 = 3\cdot 4 - 4\cdot 1 = 8 = 2\,Q_1^{(1)}.\end{aligned}$$

By uniqueness: $P_n^{(1)} = L(P^{(0)})_n$ and $2\,Q_n^{(1)} = L(Q^{(0)})_n$.

Step 2: Poincaré growth analysis. The recurrence has Poincaré characteristic roots $\lambda = 2$ and $\lambda = 1$ (from $\lambda^2 - 3\lambda + 2 = 0$). Both $P_n^{(m)}$ and $Q_n^{(m)}$ are dominant, growing like $(2n)!!$, so $P_n^{(m)}/Q_n^{(m)} \to S^{(m)}$ at rate $O(n!/(2n)!!) = O(2^{-n}/\sqrt{n})$.

Step 3: The limit. From Step 1:

$$S_n^{(1)} = 2\cdot\frac{(n{+}2)P_n^{(0)} - (n{+}1)^2 P_{n-1}^{(0)}}{(n{+}2)Q_n^{(0)} - (n{+}1)^2 Q_{n-1}^{(0)}}.$$

Dividing by $(n{+}2)Q_n^{(0)}$ and using $(n{+}1)^2/[(n{+}2)(2n{+}1)] \to \frac{1}{2}$, both correction terms converge to $\frac{1}{2}$, giving $S^{(1)} = 2\,S^{(0)}$.

Step 4: General m and the Gamma identity. The base case $S^{(0)} = 2/\pi$ is established by Theorem **??**. The Wallis reduction formula gives $I_{m+1} = \frac{2m+1}{2m+2}\,I_m$ where $I_m = \int_0^{\pi/2} \sin^{2m}(x)\,dx = \frac{\sqrt{\pi}\,\Gamma(m+\frac{1}{2})}{2\,\Gamma(m+1)}$. Assuming Conjecture **??**, the ratio formula $\mathrm{val}(m{+}1)/\,\mathrm{val}(m) = 2(m{+}1)/(2m{+}1) = I_m/I_{m+1}$ propagates inductively from $S^{(0)} = 1/I_0$, yielding $\mathrm{val}(m) = 1/I_m = 2\,\Gamma(m{+}1)/(\sqrt{\pi}\,\Gamma(m{+}\frac{1}{2}))$ for all non-negative integers m. Under the same assumption, the identity extends to all real $m > -\frac{1}{2}$ by analytic continuation. □

Remark 5.3. For $m = 0$ and $m = 1$, Steps 1–3 give a complete algebraic proof independent of Conjecture **??**: $S^{(0)} = 2/\pi$ and $S^{(1)} = 4/\pi$. For $m \geq 2$, the proof is conditional on the ratio conjecture. No polynomial-coefficient shift operator L_m mapping $\mathcal{S}_m$ to $\mathcal{S}_{m+1}$ has been found for general m [**?**].

Corollary 5.4. *Assuming Conjecture* **??**, *the following specializations hold.*

(i) *Integer $m \geq 0$:* $\mathrm{val}(m) = 2^{2m+1}/\big(\pi\binom{2m}{m}\big)$.

(ii) *Half-integer $m = k{+}\frac{1}{2}$, $k \geq 0$:* $\mathrm{val}(m)$ *is an exact rational (Wallis product partial convergent).*

(iii) $\mathrm{val}(m) \sim \sqrt{4m/\pi}$ *as $m \to \infty$ by Stirling's formula.*

6 Casoratian Series Identity for $\pi/4$

We give a Casoratian derivation of the series identity for $\pi/4$. The telescoping argument is internal to the $m = 1$ recurrence, while the identification of the continued-fraction limit as $4/\pi$ comes independently from the Euler/Wallis analysis of the previous section.

6.1 Convergence

Proposition 6.1. *The CF* (**??**) *with $\alpha(n) = -n(2n{-}3)$, $\beta(n) = 3n{+}1$ converges.*

Proof. By the Śleszyński–Pringsheim theorem, the CF converges if $|\alpha(n)| \leq \beta(n)\,\beta(n{-}1)$ for $n \geq 2$. We have $\beta(n)\,\beta(n{-}1) - |\alpha(n)| = (3n{+}1)(3n{-}2) - n(2n{-}3) = 7n^2 - 2$, which is strictly positive for $n \geq 1$. □

6.2 Casoratian recursion

Let $h_n = n^2{+}3n{+}1$ (the polynomial solution from Theorem **??**) and $g_n = q_n/(2n{-}1)!!$, both satisfying the reduced recurrence (**??**).

Lemma 6.2 (Casoratian recursion)**.** *The Casoratian $W_n = h_n\,g_{n-1} - h_{n-1}\,g_n$ satisfies $W_n = \frac{n}{2n-1}\,W_{n-1}$ for $n \geq 2$.*

Proof. Substituting the recurrence (**??**) for h_n and g_n:

$$\begin{aligned} W_n &= \frac{(3n{+}1)h_{n-1} - n\,h_{n-2}}{2n{-}1}\,g_{n-1} - h_{n-1}\,\frac{(3n{+}1)g_{n-1} - n\,g_{n-2}}{2n{-}1} \\ &= \frac{n}{2n{-}1}(h_{n-1}\,g_{n-2} - h_{n-2}\,g_{n-1}) = \frac{n}{2n{-}1}\,W_{n-1}. \end{aligned}$$ □

Theorem 6.3 (Casoratian closed form)**.** *For all $n \geq 1$, $W_n = n!/(2n{-}1)!!$.*

Proof. Base: $W_1 = 5 \cdot 1 - 1 \cdot 4 = 1 = 1!/1!!$. Induction: $W_n = \frac{n}{2n-1} \cdot \frac{(n-1)!}{(2n-3)!!} = \frac{n!}{(2n-1)!!}$. □

6.3 Main series identity

Theorem 6.4. *Let S be the CF with $\alpha(n) = -n(2n-3)$, $\beta(n) = 3n+1$.*

(i) *S converges, with $|S_n - S| = O(2^{-n}/n^{7/2})$.*

(ii) *$p_n = (2n-1)!!\,(n^2+3n+1)$.*

(iii) *$S = 4/\pi$, equivalently*

$$\frac{\pi}{4} = 1 - \sum_{k=1}^{\infty} \frac{k!}{(2k-1)!!\,(k^2+3k+1)(k^2+k-1)} = 1 - \sum_{k=1}^{\infty} \frac{2^k}{\binom{2k}{k}(k^2+3k+1)(k^2+k-1)}. \tag{8}$$

Proof of (iii). The ratio g_n/h_n telescopes:

$$\frac{g_n}{h_n} - \frac{g_{n-1}}{h_{n-1}} = \frac{-W_n}{h_n\,h_{n-1}}.$$

Summing from $k = 1$ to n and using $g_0/h_0 = 1$:

$$\frac{g_n}{h_n} = 1 - \sum_{k=1}^{n} \frac{k!}{(2k-1)!!\,(k^2+3k+1)(k^2+k-1)}.$$

As $n \to \infty$, the convergents $p_n/q_n = h_n/g_n$ tend to the continued-fraction value S by Proposition **??**. The value $S = 4/\pi$ is established independently by Lemma **??** and Step 3 of the proof of Theorem **??**: the raising operator L maps $P^{(0)}, Q^{(0)}$ to scalar multiples of $P^{(1)}, Q^{(1)}$, so $S^{(1)} = 2\,S^{(0)} = 4/\pi$ by Theorem **??**, with no appeal to the present section. Therefore $g_n/h_n \to \pi/4$, and passing to the limit in the telescoping identity yields (**??**). The alternative form uses $k!/(2k-1)!! = 2^k/\binom{2k}{k}$ (see [**?**, §1.2.6] for the double-factorial convention). □

Proof of convergence rate (i). By Stirling, $\binom{2k}{k} \sim 4^k/\sqrt{\pi k}$, so the general term satisfies $t_k \sim \sqrt{\pi k}/(2^k k^4) = O(2^{-k}/k^{7/2})$. Since the tail is dominated by its first term, $|S_N - \pi/4| = O(2^{-N}/N^{7/2})$. At $N = 35$ the error drops below 10^{-16}. □

7 Gauss CF Non-Membership

7.1 Factorisation over $\mathbb{Q}(\sqrt{5})$

Set $\varphi = (1+\sqrt{5})/2$, $\psi = (1-\sqrt{5})/2$. Since $(k+\varphi)(k+\psi) = k^2+k-1$ and $(k+2+\varphi)(k+2+\psi) = k^2+5k+5$, the ratio of consecutive series terms in (**??**) is

$$\frac{t_{k+1}}{t_k} = \frac{(k+1)(k+\varphi)(k+\psi)}{2(k+\frac{1}{2})(k+2+\varphi)(k+2+\psi)}. \tag{9}$$

7.2 Pochhammer obstruction

A standard ${}_pF_q\left(\begin{matrix}\cdots\\ \cdots\end{matrix}\middle| z\right)$ has term ratio equal to a product of linear factors $(k+a_i)/(k+b_j)$ with $a_i, b_j \in \mathbb{Q}$. Here the irreducible quadratics k^2+k-1 and k^2+5k+5 (both of discriminant 5) force the parameters into $\mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}$.

7.3 Gauss CF non-membership

Proposition 7.1. *The CF does not arise from any ${}_2F_1$ ratio via the Gauss continued-fraction theorem.*

Proof. The Gauss CF for ${}_2F_1\left(\begin{smallmatrix} A+1, B \\ C+1 \end{smallmatrix}\middle| z\right) / {}_2F_1\left(\begin{smallmatrix} A, B \\ C \end{smallmatrix}\middle| z\right)$ has unit-denominator coefficients t_k determined by $(A, B, C) \in \mathbb{Q}^3$. Solving $t_1 = c_1$, $t_2 = c_2$, $t_3 = c_3$ (where the c_k are the equivalence-transformed coefficients of our CF) yields a degree-6 polynomial system with six solutions; four are complex and two are real with $C \approx -3.259$. For these, $|t_4 - c_4| \approx 5.9$, so no match exists. □

7.4 ${}_2F_3$ identification

Remark 7.2. The term ratio (**??**) identifies the series as

$$ {}_2F_3\big(\varphi, \psi;\ \tfrac{1}{2},\ 2{+}\varphi,\ 2{+}\psi;\ \tfrac{1}{2}\big) $$

over $\mathbb{Q}(\sqrt{5})$. Since both the parameters and the function class depend on the extension $\mathbb{Q}(\sqrt{5})/\mathbb{Q}$, the series does not reduce to any standard ${}_pF_q$ with rational parameters.

8 Euler CF Duality

The $m = 0$ CF decomposes as $S^{(0)} = \beta(0) + \alpha(1)/T = 1 - 1/T$ where T is the inner tail starting at $n = 1$. Since $S^{(0)} = 2/\pi$ (Theorem **??**):

Proposition 8.1. $T = \pi/(\pi - 2)$, *equivalently* $T/(T{-}1) = \pi/2$.

Proof. $T = 1/(1 - S^{(0)}) = 1/(1 - 2/\pi) = \pi/(\pi - 2)$. The identity $T/(T{-}1) = \pi/2$ then follows from $T - 1 = (2)/(\pi - 2)$. □

Remark 8.2. This connects our PCF to the classical identity $\pi/2 = \sum_{k=0}^{\infty} k!/(2k{+}1)!!$ from $\arcsin(1)$ [**?**, §4.24.2]. The partial numerators of the $m = 0$ and $m = 1$ families differ by a linear term: $\alpha_0(n) - \alpha_1(n) = -n(2n{-}1) + n(2n{-}3) = -2n$.

9 Comparison with Brouncker's CF

Lord Brouncker's classical CF for $4/\pi$ is

$$ \frac{4}{\pi} = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cdots}}}, $$

with $\alpha^B(n) = (2n{-}1)^2$ and constant $\beta^B(n) = 2$. Both CFs converge to $4/\pi$ but differ structurally:

	Brouncker	**Novel** ($m{=}1$)
$\alpha(n)$	$(2n{-}1)^2$	$-n(2n{-}3)$
$\beta(n)$	2 (constant)	$3n{+}1$ (linear)
p_n	$(2n{+}1)!!$	$(2n{-}1)!!(n^2{+}3n{+}1)$
Series	Leibniz $\sum(-1)^k/(2k{+}1)$	Eq. (**??**)
Rate	$O(1/n)$	$O(2^{-n})$

	Brouncker			Novel		
n	p_n	q_n	S_n	p_n	q_n	S_n
0	1	1	1.000000	1	1	1.000000
1	3	2	1.500000	5	4	1.250000
2	15	13	1.153846	33	26	1.269231
3	105	76	1.381579	285	224	1.272321
4	945	789	1.197719	3045	2392	1.272993
5	10395	7734	1.344065	38745	30432	1.273166
8	34459425	28018665	1.229874	180405225	141690240	1.273237
						$4/\pi \approx 1.273\,239\,545$

At $n = 8$, Brouncker's error is $\sim 4 \times 10^{-2}$ while the novel CF error is $\sim 3 \times 10^{-6}$, a factor of 10^4. The two CFs are not related by any equivalence transformation, contraction, or Möbius substitution.

10 Convergence Theory

10.1 Worpitzky condition

Lemma 10.1. *For both families, the normalised partial numerators $|\tilde{\alpha}_n| = |\alpha_n|/(\beta_n\,\beta_{n-1})$ satisfy $|\tilde{\alpha}_n| < 1/4$ for all sufficiently large n.*

Proof. **Log family.** $|\tilde{\alpha}_n| \to k/(k{+}1)^2 < 1/4$ for $k > 1$ (since $(k{-}1)^2 > 0$).

Pi family. $|\tilde{\alpha}_n| = n(2n{-}1)/((3n{+}1)(3n{-}2)) \to 2/9 < 1/4$. At $n = 1$: $|\tilde{\alpha}_1| = 1/4$ (borderline); strict inequality holds for $n \geq 2$, and convergence at equality at finitely many n follows from the classical extension [**?**, §4.2]. □

10.2 Bracketing

Lemma 10.2. *If $\alpha_n < 0$ and $\beta_n > 0$ for all $n \geq 1$, and the CF converges to L, then* $\min(C_N, C_{N+1}) \leq L \leq \max(C_N, C_{N+1})$.

Proof. The determinant $p_n q_{n-1} - p_{n-1} q_n = \prod_{j=1}^{n}(-\alpha_j) > 0$ shows $C_n - C_{n-1}$ alternates in sign. See [**?**, Theorem 4.35]. □

11 Meta-Family

The unified template $\alpha(n) = -(\alpha_0 n^2 + \beta_0 n)$, $\beta(n) = \gamma_0 n + \delta_0$ contains both the Logarithmic Ladder ($\beta_0 = 0$, $\delta_0 = \alpha_0$, $\gamma_0 = \alpha_0 + 1$) and the Pi Family ($\alpha_0 = 2$, $\gamma_0 = 3$, $\delta_0 = 1$) as special cases. Two additional sub-families were discovered and verified to 80 digits:

- $(\alpha_0, \beta_0, \gamma_0, \delta_0) = (1, 0, 4, 2) \to 2/\ln 3$;
- $(4, 0, 8, 4) \to 4/\ln 3$.

Both are instances of the Logarithmic Ladder at $k = 3$ (and its equivalence-transformed double).

12 Quadratic GCF Constants

Define $V(A, B, C) = 1 + \mathbf{K}_{n\geq 1}\, 1/(An^2 + Bn + C)$.

Proposition 12.1. *For all (A, B, C) with $A \geq 1$, $C \geq 1$, and $An^2 + Bn + C > 0$ for $n \geq 1$, $V(A, B, C)$ is irrational.*

Proof. The convergent denominators satisfy $Q_n = (An^2+Bn+C)\,Q_{n-1} + Q_{n-2}$ with $\log Q_n \gtrsim n\log n$. The Wronskian $|P_nQ_{n-1} - P_{n-1}Q_n| = 1$ for all n. By Euler's irrationality criterion, the limit is irrational. □

Our systematic scan found **482 distinct** such constants. A PSLQ [**?**] integer relation search at 300-digit precision over 15 standard constants $\{\pi, \pi^2, e, \ln 2, \ln 3, \gamma, G, \zeta(3), \zeta(5), \sqrt{2}, \sqrt{3}, \sqrt{5}, \varphi, \sqrt{\pi}, \pi\ln 2\}$ finds no relation for any V-constant, and none is algebraic of degree ≤ 8 with coefficients $\leq 10^5$. Pairwise cross-relation tests on 91 representative constants (435 pairs, PSLQ with coefficients up to 500) and triple-relation tests (455 triples, coefficients up to 200) all return negative. A targeted sweep against 588 expressions built from Gamma ratios and Pi Family values also produces no matches.

The only structural relation found is the *shift symmetry* $V(1,B,C) - V(1,-B,C) \in \mathbb{Z}$ for all tested B, C, following from the palindromic structure of the GCF denominators. This gives strong (but non-rigorous) evidence of mutual algebraic independence.

Remark 12.2. An earlier PSLQ search returned $[0, 1, -2, 1]$ relative to $\{V, \sqrt{5}, \varphi, 1\}$; this is the trivial identity $\sqrt{5} - 2\varphi + 1 = 0$ with zero coefficient on V.

13 Numerical Certification

All results are certified using Arb ball arithmetic (`python-flint` 0.8.0, 10 000-bit precision) at depth $N = 5000$. By Lemma **??**, the limit lies in $[\min(C_N, C_{N-1}), \max(C_N, C_{N-1})]$.

Table 4: Arb-certified intervals (depth 5000)

PCF	Target	Bracket width	Digits
$\alpha{=}{-}2n^2,\ \beta{=}3n{+}2$	$1/\ln 2$	$< 10^{-1509}$	1509
$\alpha{=}{-}3n^2,\ \beta{=}4n{+}3$	$1/\ln(3/2)$	$< 10^{-2265}$	2265
$\alpha{=}{-}n(2n{-}1),\ \beta{=}3n{+}1$	$2/\pi$	$< 10^{-1508}$	1508
$\alpha{=}{-}n(2n{-}3),\ \beta{=}3n{+}1$	$4/\pi$	$< 10^{-1518}$	1518

For the $m = 1$ CF, backward recurrence at depth 500 with 80-digit arithmetic confirms

$$S = 1.273239544735162686151070106980114896275677165923651589 98\ldots$$

matching $4/\pi$ to all 60 digits.

14 Discovery Methodology

1. **MITM sweep.** Fix $\beta(n) = dn{+}e$; enumerate $\alpha(n) = pn^2{+}qn{+}r$ with $|p|, |q|, |r| \leq 5$; match PCF values against a database of 200 known constants.

2. **Modular signature sweep.** Vary $d \in \{2, \ldots, 8\}$, $e \in \{1, \ldots, 8\}$. The $d{=}3$ family produced both π- and ln-hits.

3. **Parametric generalisation.** Sweep c in $\alpha(n) = -n(2n{-}c)$ and k in $\alpha(n) = -kn^2$, discovering the full infinite families.

15 Open Questions

1. **Prove Conjecture ??** (the ratio formula for general m). A proof would complete Theorem **??** unconditionally. Clean closed forms for $p_n^{(m)}$ and $q_n^{(m)}$ for general $m > 0$ would likely suffice.

2. **Irrationality measures.** Do the convergent sequences yield new irrationality measures for $\ln 2$, $\ln(3/2)$, or $1/\pi$ via Apéry-style arguments?

3. **Higher denominators.** Do analogous families exist for $d \geq 5$, producing $\zeta(3)$, Catalan's constant, or polylogarithms?

4. **Quadratic GCFs.** Determine whether $V(3,1,1) \approx 1.19737399068835760\ldots$ has any closed form. This constant resists 3 400+ PSLQ tests against 15 special-function families [**?**].

5. **Algebraic independence.** Prove (or disprove) that the 482 quadratic GCF constants are mutually algebraically independent.

A Exact Convergents

Log (k=2): $p_n = (n{+}1)! \cdot 2^{n+1}$.

n	p_n	q_n	C_n
0	2	1	2.000000
1	8	5	1.600000
2	48	32	1.500000
3	384	262	1.465649
4	3840	2644	1.452345
5	46080	31832	1.447584

Limit: $1/\ln 2 = 1.442695\ldots$

Pi (m=0): $p_n = (2n{+}1)!!$.

n	p_n	q_n	C_n
0	1	1	1.000000
1	3	4	0.750000
2	15	22	0.681818
3	105	160	0.656250
4	945	1464	0.645492
5	10395	16224	0.640688

Limit: $2/\pi = 0.636620\ldots$

Pi (m=1): $p_n = (2n{-}1)!!\,(n^2{+}3n{+}1)$.

n	p_n	q_n
0	1	1
1	5	4
2	33	26
3	285	224
4	3 045	2 392
5	38 745	30 432
6	571 725	449 040
7	9 594 585	7 535 616
8	180 405 225	141 690 240

Limit: $4/\pi = 1.273239\ldots$

B Reproducibility

Python 3.12, `mpmath` 1.4.1, `python-flint` 0.8.0 (Arb), `sympy` 1.14.0. Forward recurrence (**??**) with p_{-1}=1, p_0=$\beta(0)$, q_{-1}=0, q_0=1. Runtime: <2 hours on a standard desktop.

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References

[1] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$*, Astérisque **61** (1979), 11–13.

[2] A. Cuyt, V. B. Petersen, B. Verdonk, H. Waadeland, and W. B. Jones, *Handbook of Continued Fractions for Special Functions*, Springer, 2008.

[3] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, et al. (eds.), *NIST Digital Library of Mathematical Functions*, Release 1.2.3, `https://dlmf.nist.gov/`, 2024.

[4] D. Elimelech et al., *Algorithm-assisted discovery of an intrinsic order among mathematical constants*, Proc. Natl. Acad. Sci. USA **121**(25) (2024), e2321440121.

[5] H. R. P. Ferguson, D. H. Bailey, and S. Arwade, *Analysis of PSLQ, an integer relation finding algorithm*, Math. Comp. **68** (1999), 351–369.

[6] D. E. Knuth, *The Art of Computer Programming*, vol. 1, 3rd ed., Addison-Wesley, 1997.

[7] L. Lorentzen and H. Waadeland, *Continued Fractions with Applications*, 2nd ed., Atlantis Press, 2008.

[8] OEIS Foundation, *A028387: $n(n{+}2)+1$*, The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A028387`.

[9] G. Raayoni, S. Gottlieb, Y. Manor, G. Pisha, Y. Harris, U. Mendlovic, D. Haviv, Y. Hadad, and I. Kaminer, *Generating conjectures on fundamental constants with the Ramanujan Machine*, Nature **590** (2021), 67–73.

[10] H. S. Wall, *Analytic Theory of Continued Fractions*, Van Nostrand, 1948; reprinted Chelsea, 1967.